

VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN
(Course Code: ECE 1003)

Module-2:Lecture-7

Combinational Circuit Design

CONTENTS

Module-2

- ❖ Binary adder and subtractor
- ❖ Ripple carry adders/subtractors and fast adders
- ❖ Binary decoders, encoders
- ❖ Multiplexers and de-multiplexers
- ❖ Logic functions using decoders and multiplexers
- ❖ **Code converters**
- ❖ Magnitude comparator

CONTENTS

Lecture-7

- ❖ BCD to 7 segment display
- ❖ BCD to Excess-3 Code
- ❖ Excess-3 to BCD code

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

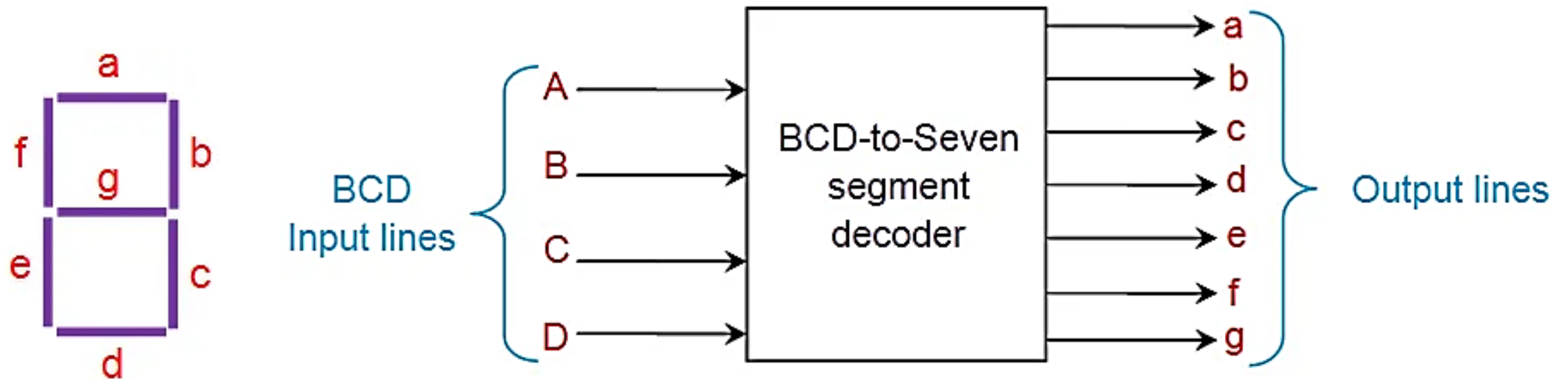
References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Code Converter

BCD TO 7-SEGMENT CONVERSION

- ✓ **BCD** (Binary Coded Decimal) is an encoding scheme which represents each of the decimal numbers by its equivalent **4-bit binary pattern**.
- ✓ **Seven segment** displays comprise of seven individual segments formed by either Light Emitting Diodes (LEDs) or Liquid Crystal Displays (LCDs) arranged in a definite pattern.



BCD TO 7-SEGMENT CONVERSION

Step-1:

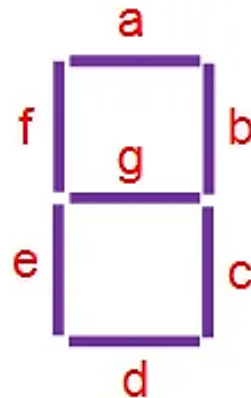
Number of BCD inputs = 4

Number of 7-segment output = 7

Step- 2:

The truth table can be obtained as

From the truth table, it can be observed that BCD takes 4-bit binary bits which is from 0000 to 1001. However, the binary inputs with combinations **1010 to 1111** are **unused**, so the corresponding output will be considered as **don't care conditions**

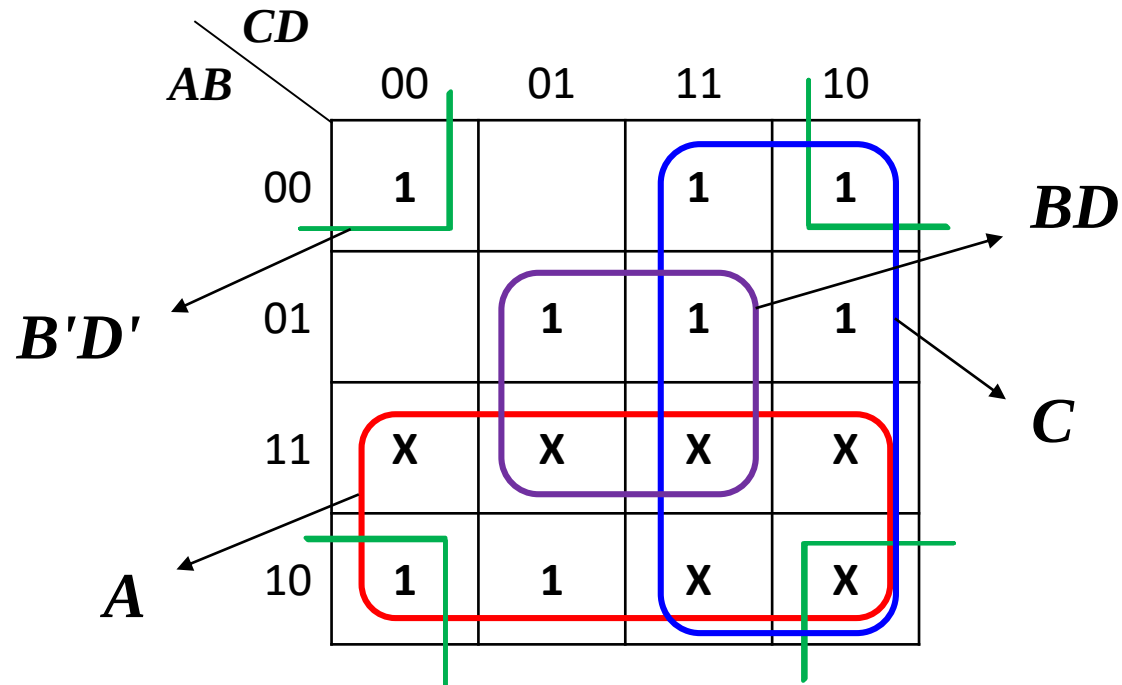


Decimal Digit	Input lines				Output lines							Display pattern
	A	B	C	D	a	b	c	d	e	f	g	
0	0	0	0	0	1	1	1	1	1	1	0	0
1	0	0	0	1	0	1	1	0	0	0	0	1
2	0	0	1	0	1	1	0	1	1	0	1	2
3	0	0	1	1	1	1	1	1	0	0	1	3
4	0	1	0	0	0	1	1	0	0	1	1	4
5	0	1	0	1	1	0	1	1	0	1	1	5
6	0	1	1	0	1	0	1	1	1	1	1	6
7	0	1	1	1	1	1	1	0	0	0	0	7
8	1	0	0	0	1	1	1	1	1	1	1	8
9	1	0	0	1	1	1	1	1	0	1	1	9

BCD TO 7-SEGMENT CONVERSION

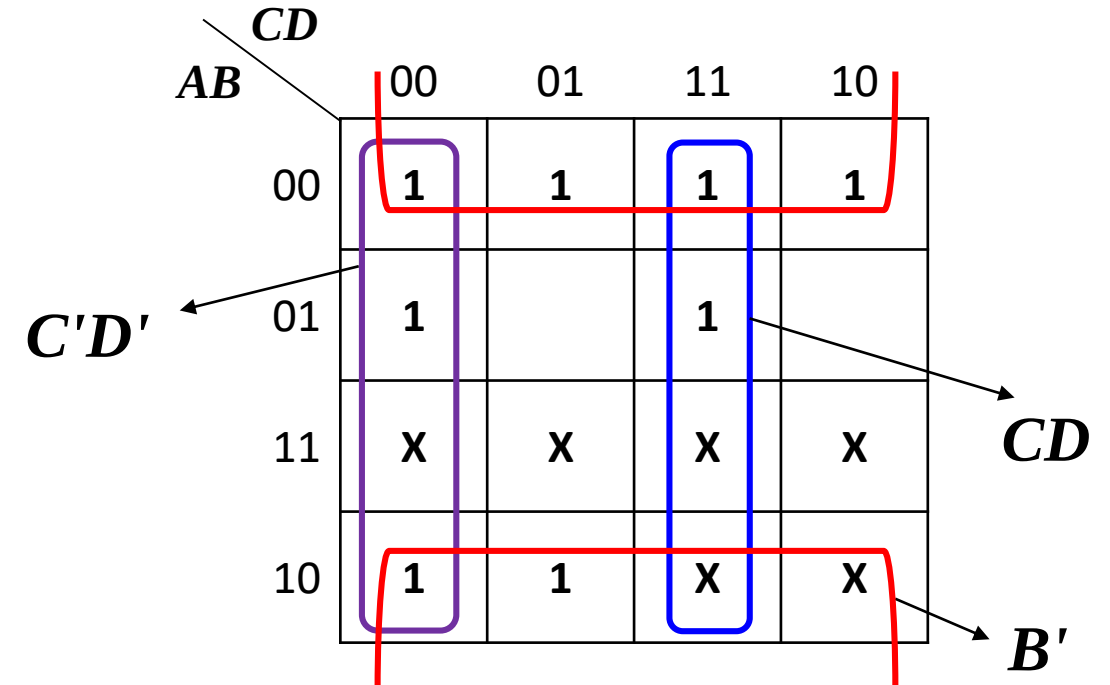
Step-3: Minimization for the output variables

For **a**



$$a = A + C + B'D' + BD$$

For **b**

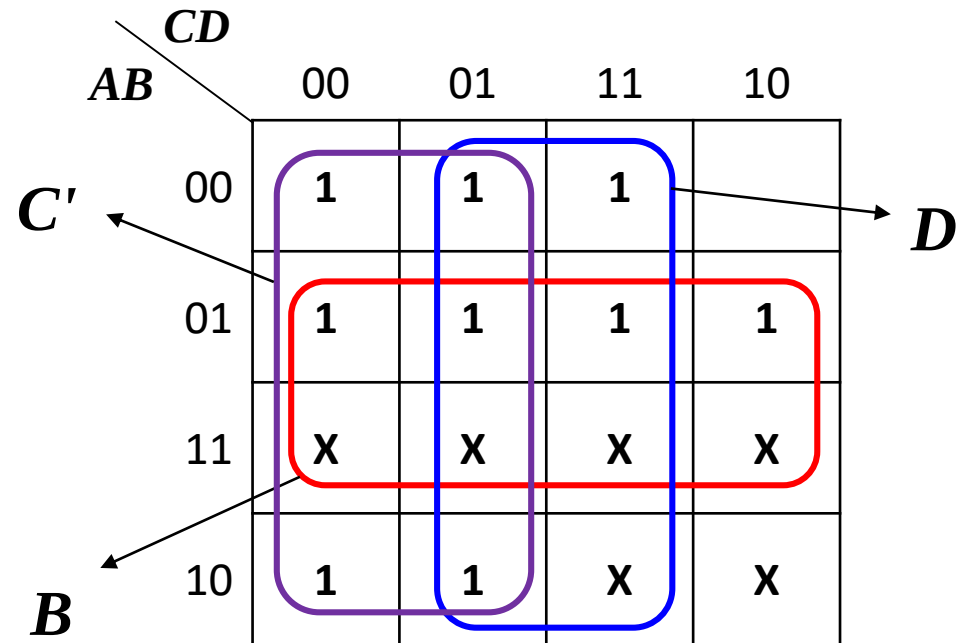


$$b = B' + C'D' + CD$$

BCD TO 7-SEGMENT CONVERSION

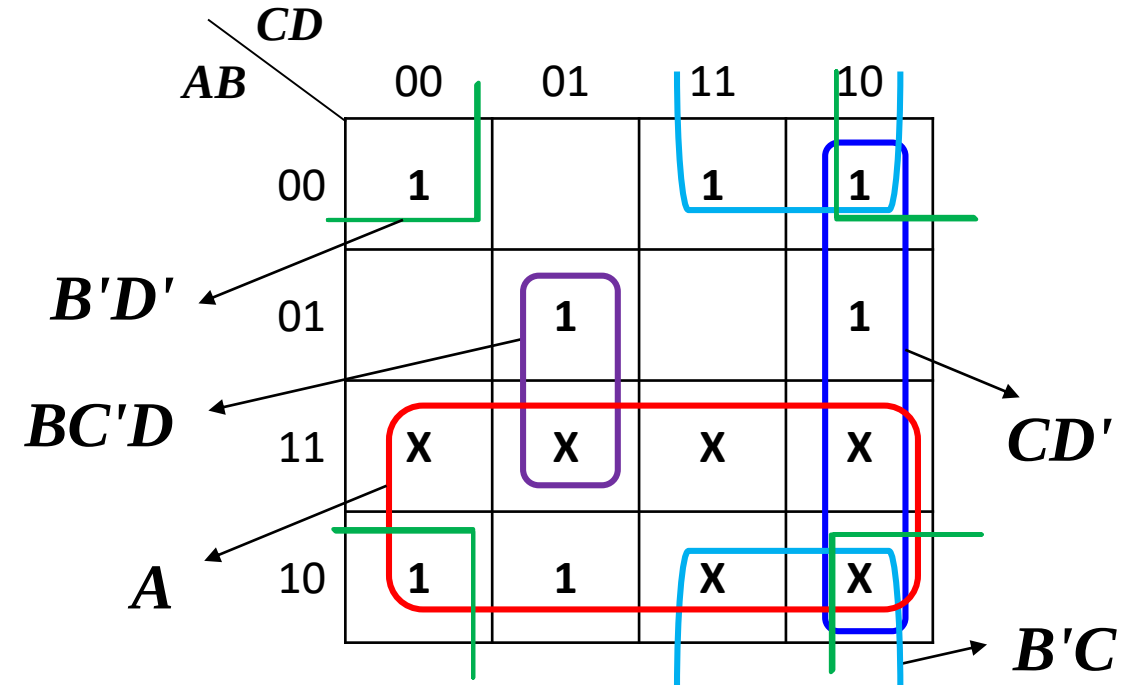
Step-3: Minimization for the output variables

For **c**



$$c = B + C' + D$$

For **d**

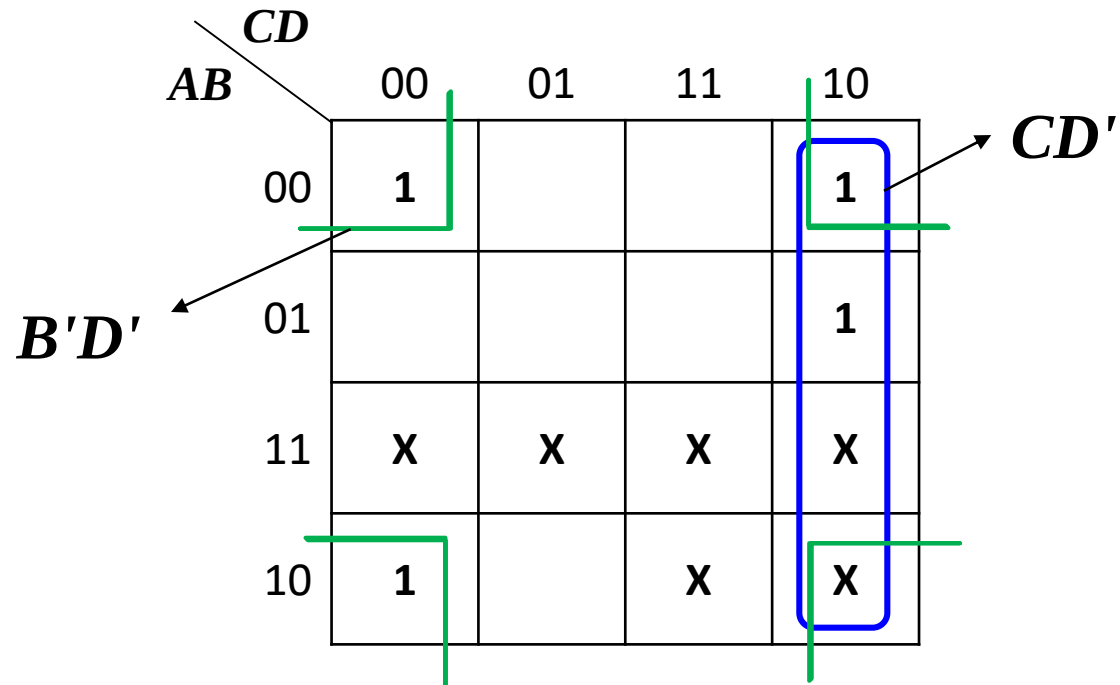


$$d = A + B'C + B'D' + CD' + BC'D$$

BCD TO 7-SEGMENT CONVERSION

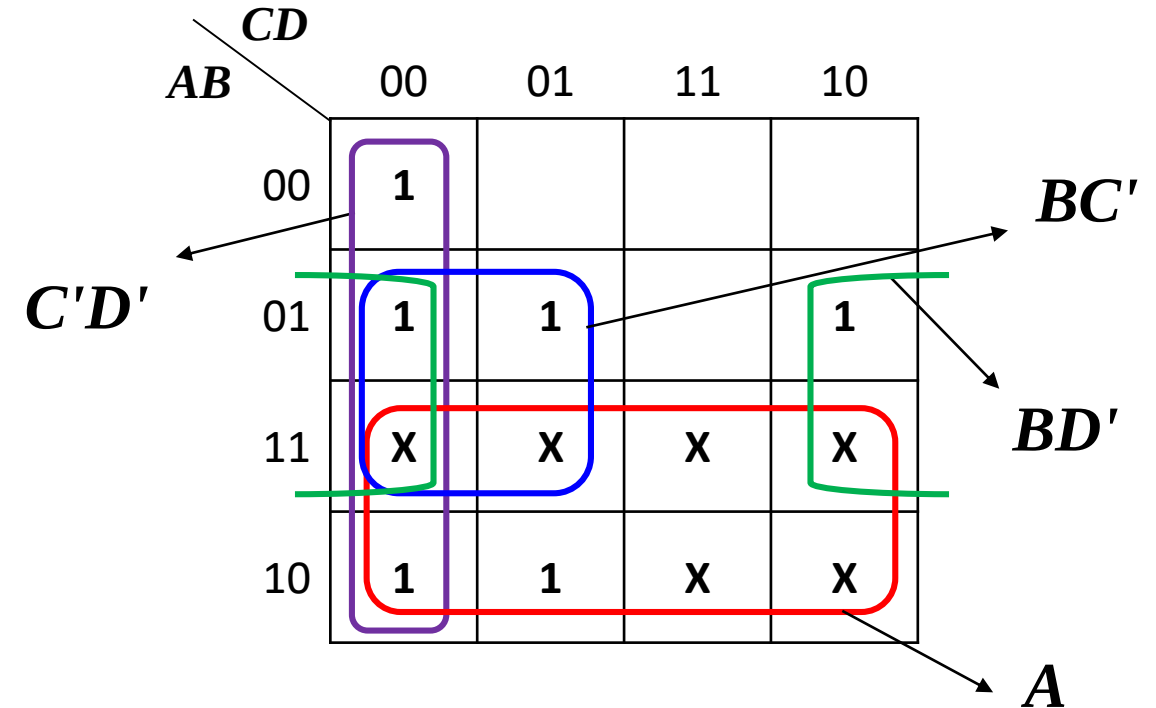
Step-3: Minimization for the output variables

For **e**



$$e = B'D' + CD'$$

For **f**

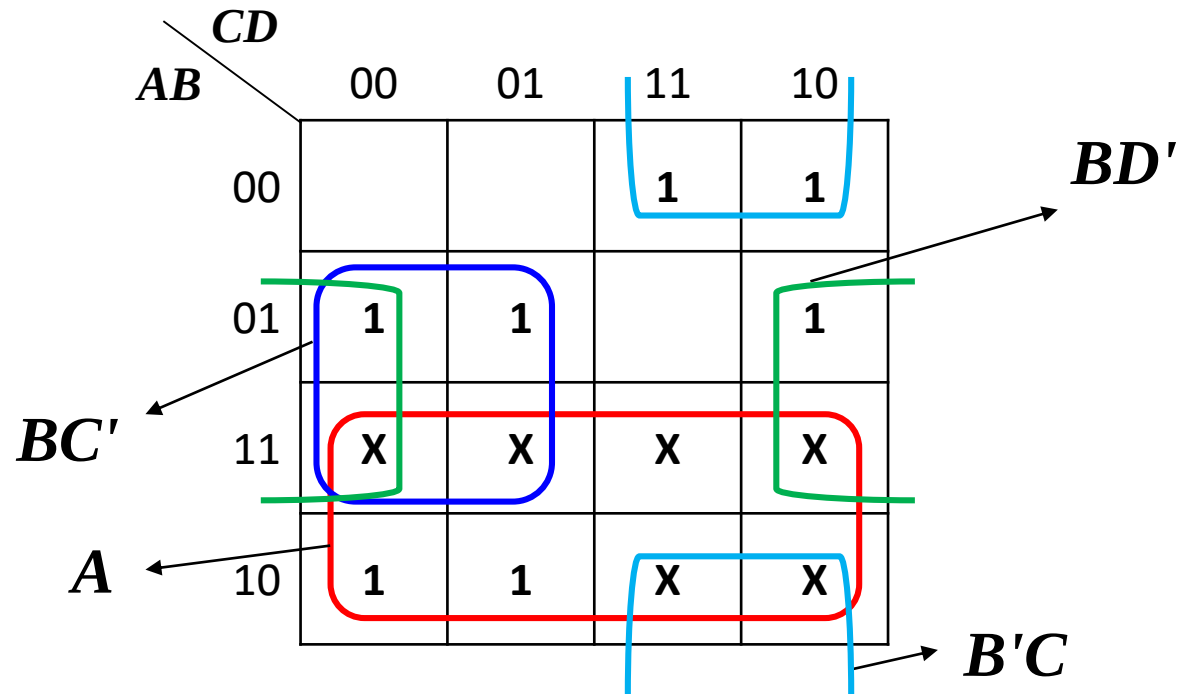


$$f = A + BC' + BD' + C'D'$$

BCD TO 7-SEGMENT CONVERSION

Step-3: Minimization for the output variables

For **g**



$$g = A + BC' + B'C + BD'$$

BCD TO 7-SEGMENT CONVERSION

Step-3: Minimization for the output variables

$$a = A + C + B'D' + BD$$

$$b = B' + C'D' + CD$$

$$c = B + C' + D$$

$$d = A + B'C + B'D' + CD' + BC'D$$

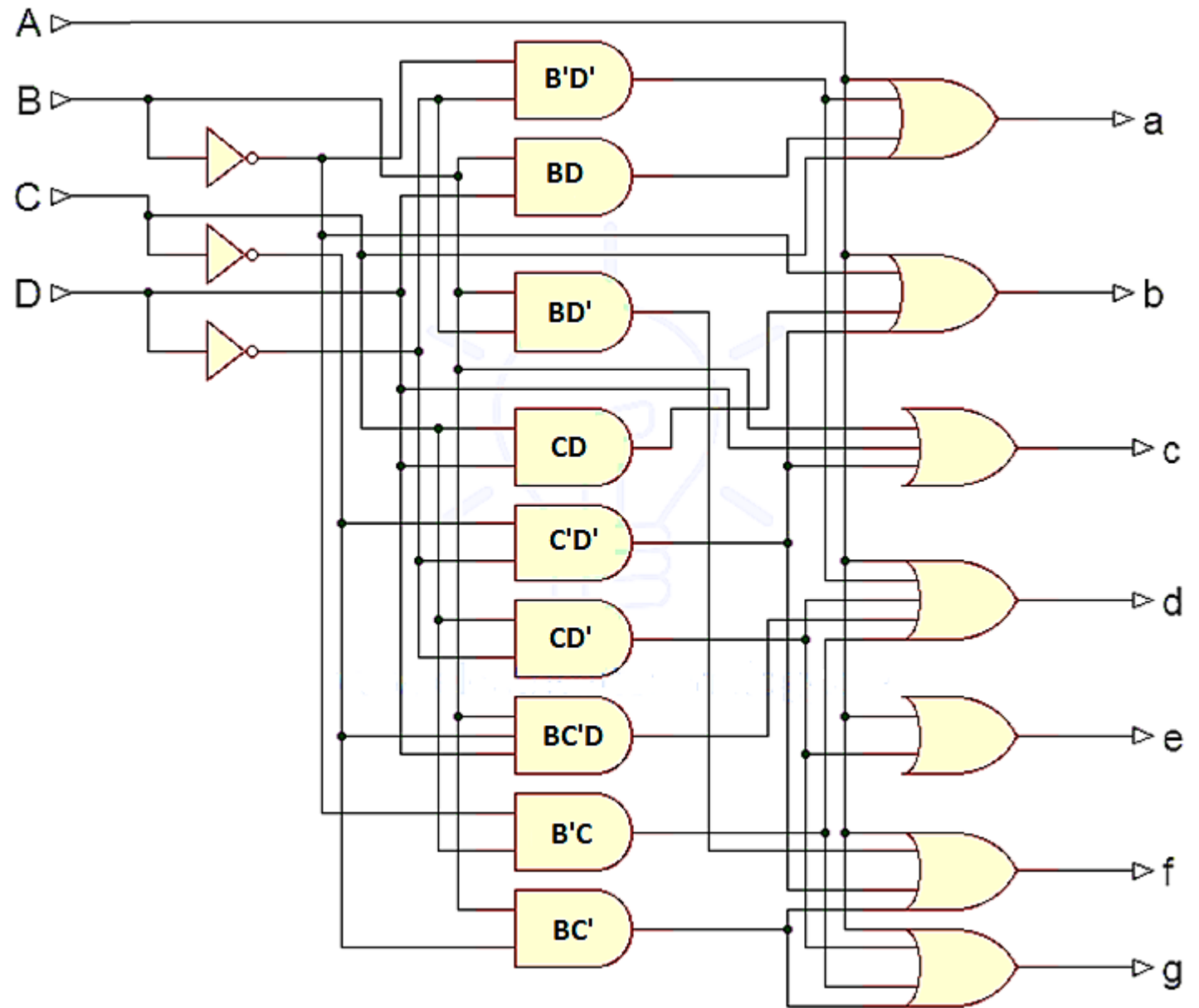
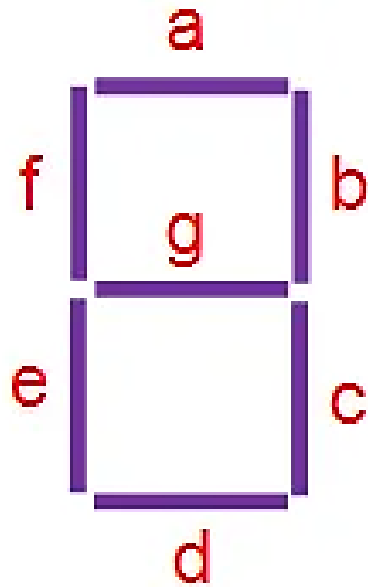
$$e = B'D' + CD'$$

$$f = A + BC' + BD' + C'D'$$

$$g = A + BC' + B'C + BD'$$

BCD TO 7-SEGMENT CONVERSION

Step-4: Circuit Design



BCD TO EXCESS-3 CODE CONVERSION

Step-1:

Number of BCD inputs = 4

Number of Excess-3 outputs = 4

Step- 2:

The truth table can be obtained as

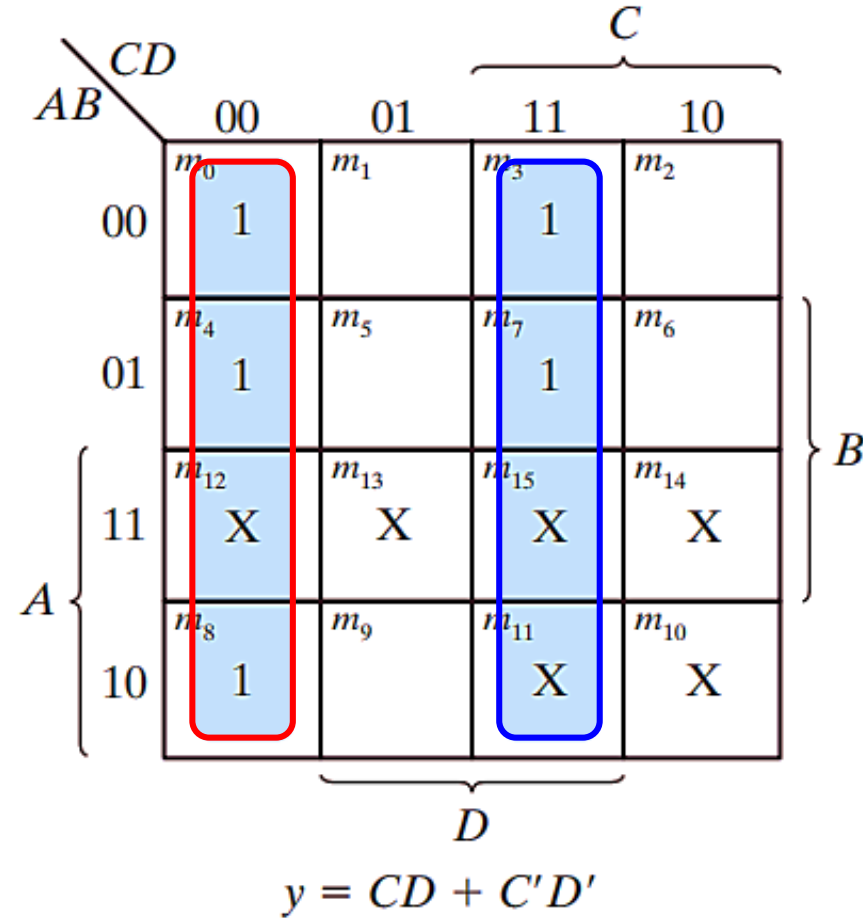
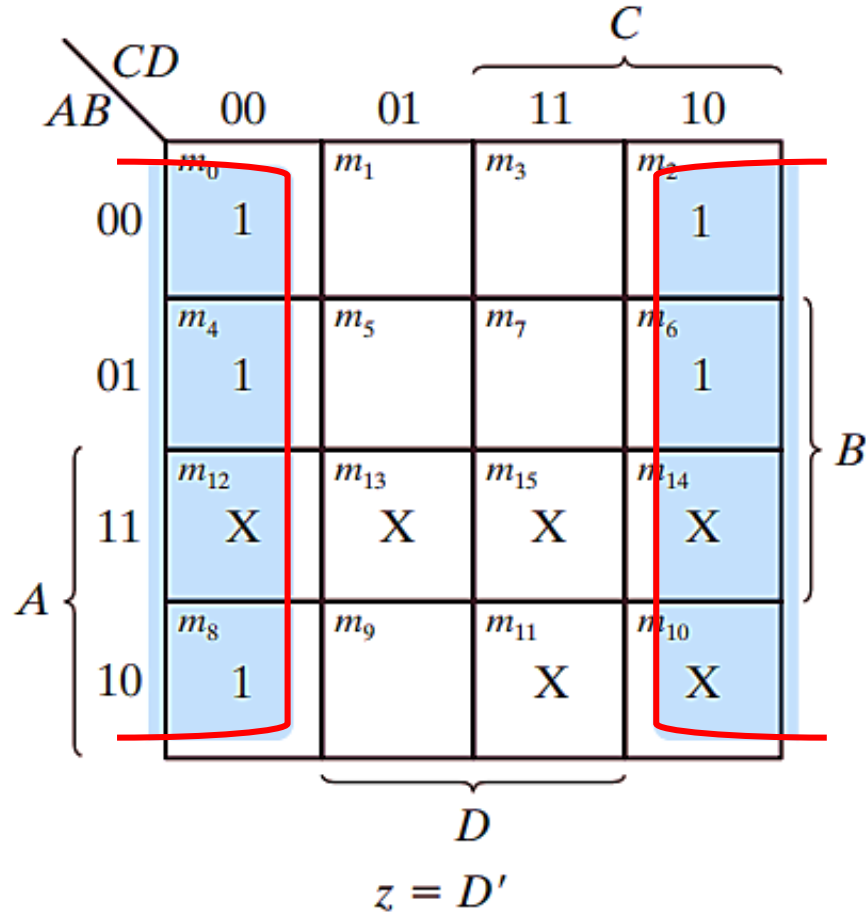
From the truth table, it can be observed that BCD takes 4-bit binary bits which is from 0000 to 1001. However, the binary inputs with combinations **1010 to 1111** are **unused**, so the corresponding output will be considered as **don't care conditions**

Truth Table for Code Conversion Example

Input BCD				Output Excess-3 Code			
A	B	C	D	w	x	y	z
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

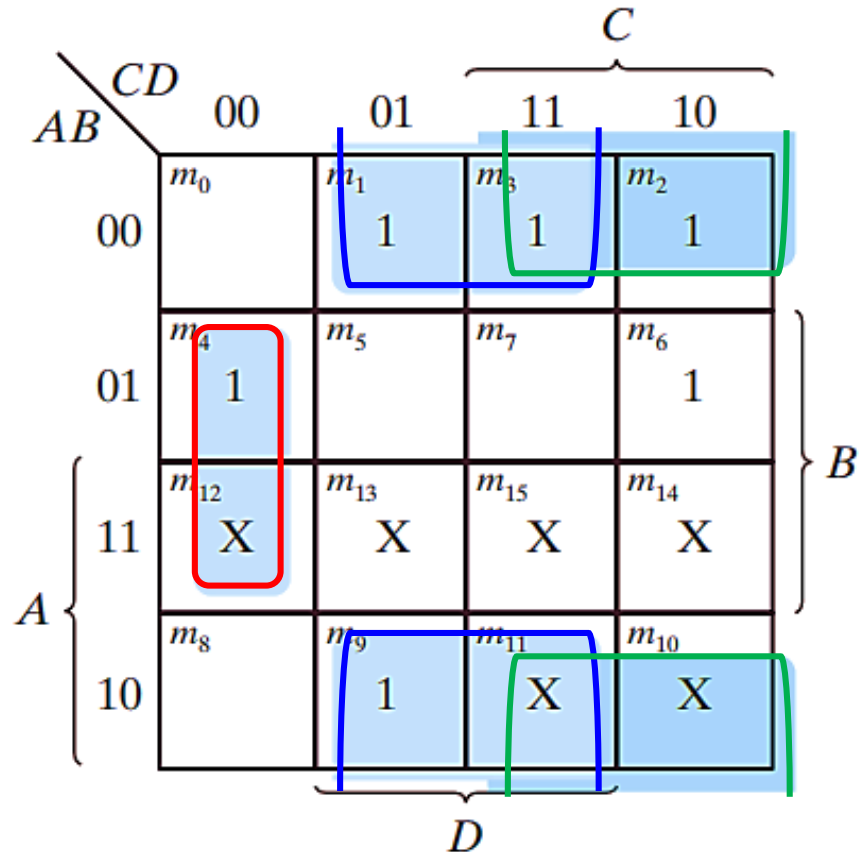
BCD TO EXCESS-3 CODE CONVERSION

Step-3: Minimization for the output variables

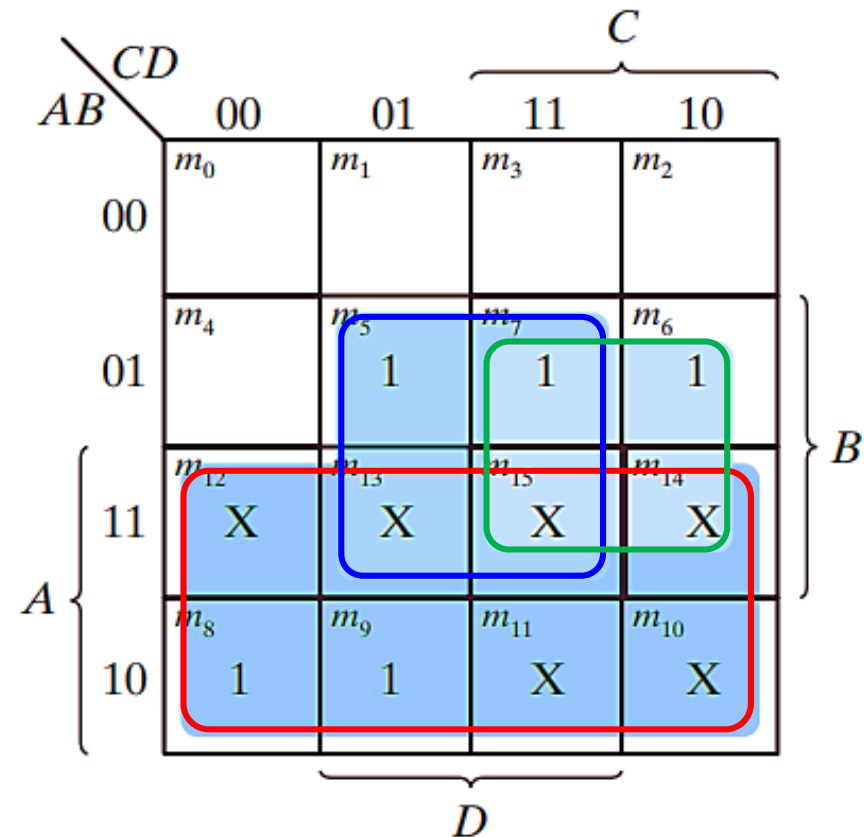


BCD TO EXCESS-3 CODE CONVERSION

Step-3: Minimization for the output variables



$$x = B'C + B'D + BC'D'$$



$$w = A + BC + BD$$

BCD TO EXCESS-3 CODE CONVERSION

Step-3: Minimization for the output variables

$$z = D'$$

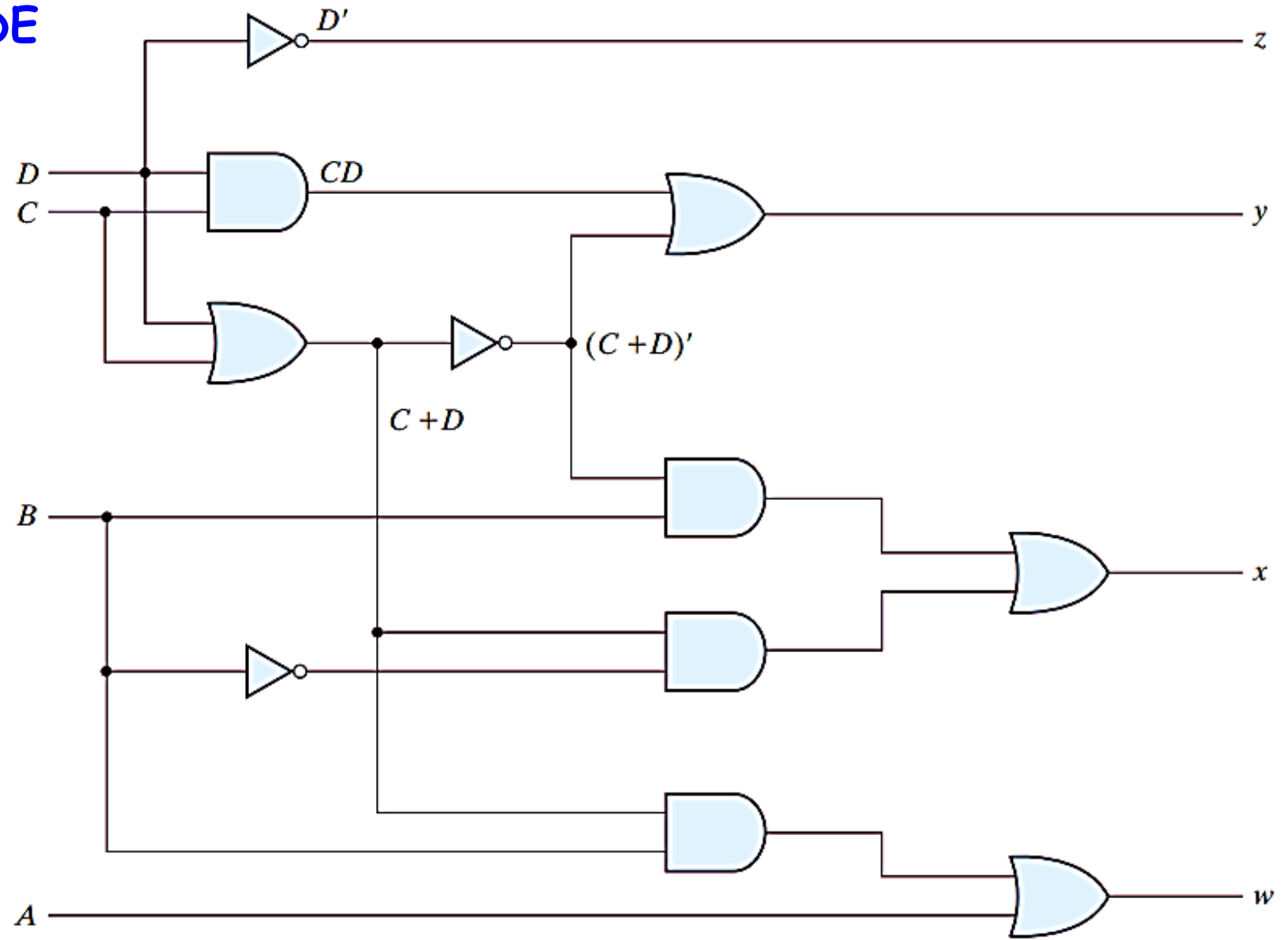
$$y = CD + C'D' = CD + (C + D)'$$

$$\begin{aligned} x &= B'C + B'D + BC'D' = B'(C + D) + BC'D' \\ &= B'(C + D) + B(C + D)' \end{aligned}$$

$$w = A + BC + BD = A + B(C + D)$$

BCD TO EXCESS-3 CODE CONVERSION

Step-4: Circuit Design



EXCESS-3 CODE TO BCD CONVERSION

Step-1:

Number of BCD inputs = 4

Number of Excess-3 outputs = 4

Step- 2:

The truth table can be obtained as

From the truth table, it can be observed
that inputs with combinations **0000**,
0001, **0010**, **1101**, **1110** and **1111** are
unused, so the corresponding output will
be considered as **don't care conditions**

	Excess-3				BCD			
	w	x	y	z	A	B	C	D
	0	0	0	0	X	X	X	X
	0	0	0	1	X	X	X	X
	0	0	1	0	X	X	X	X
3	0	0	1	1	0	0	0	0
4	0	1	0	0	0	0	0	1
5	0	1	0	1	0	0	1	0
6	0	1	1	0	0	0	1	1
7	0	1	1	1	0	1	0	0
8	1	0	0	0	0	1	0	1
9	1	0	0	1	0	1	1	0
10	1	0	1	0	0	1	1	1
11	1	0	1	1	1	0	0	0
12	1	1	0	0	1	0	0	1
	1	1	0	1	X	X	X	X
	1	1	1	0	X	X	X	X
	1	1	1	1	X	X	X	X

EXCESS-3 CODE TO BCD CONVERSION

Step-3: Minimization for the output variables

For **A**

		yz			
		00	01	11	10
wx	00	x	x		x
	01				
	11	1	x	x	x
	10			1	

Diagram illustrating the Karnaugh map for output variable A. The map shows 1s at (wx, yz) = (11, 00), (11, 01), (11, 11), and (10, 11). A red box groups the 1s at (11, 00), (11, 01), (11, 11), and (11, 10), which corresponds to the term wx . A blue box groups the 1s at (11, 11) and (10, 11), which corresponds to the term wyz .

$$A = wx + wyz$$

For **B**

		yz			
		00	01	11	10
wx	00	x	x		x
	01			1	
	11		x	x	x
	10	1	1		1

Diagram illustrating the Karnaugh map for output variable B. The map shows 1s at (wx, yz) = (00, 00), (00, 01), (01, 11), (10, 00), (10, 01), (10, 11), and (10, 10). A red box groups the 1s at (00, 00), (00, 01), (10, 00), and (10, 01), which corresponds to the term $x'y'$. A green box groups the 1s at (00, 00), (00, 01), (10, 00), and (10, 01), which corresponds to the term $x'z'$. A blue box groups the 1s at (01, 11) and (11, 11), which corresponds to the term xyz .

$$B = x'y' + x'z' + xyz$$

EXCESS-3 CODE TO BCD CONVERSION

Step-3: Minimization for the output variables

For C

		yz			
		00	01	11	10
wx	00	X	X		X
	01		1		1
	11		X	X	X
	10		1		1

Diagram illustrating the Karnaugh map for output variable C . The map shows four groups: a red vertical group covering $y'z$ (cells 0001, 0101, 1101, 1001) and a blue vertical group covering yz' (cells 0010, 0110, 1110, 1010). Arrows point from the labels $y'z$ and yz' to their respective groups.

$$C = y'z + yz'$$

For D

		yz			
		00	01	11	10
wx	00	X	X		X
	01	1			1
	11	1	X	X	X
	10	1			1

Diagram illustrating the Karnaugh map for output variable D . The map shows two red groups: a vertical group covering z' (cells 0000, 0100, 1100, 1000) and a horizontal group covering z' (cells 0000, 0100, 1100, 1000). An arrow points from the label z' to the vertical group.

$$D = z'$$

EXCESS-3 CODE TO BCD CONVERSION

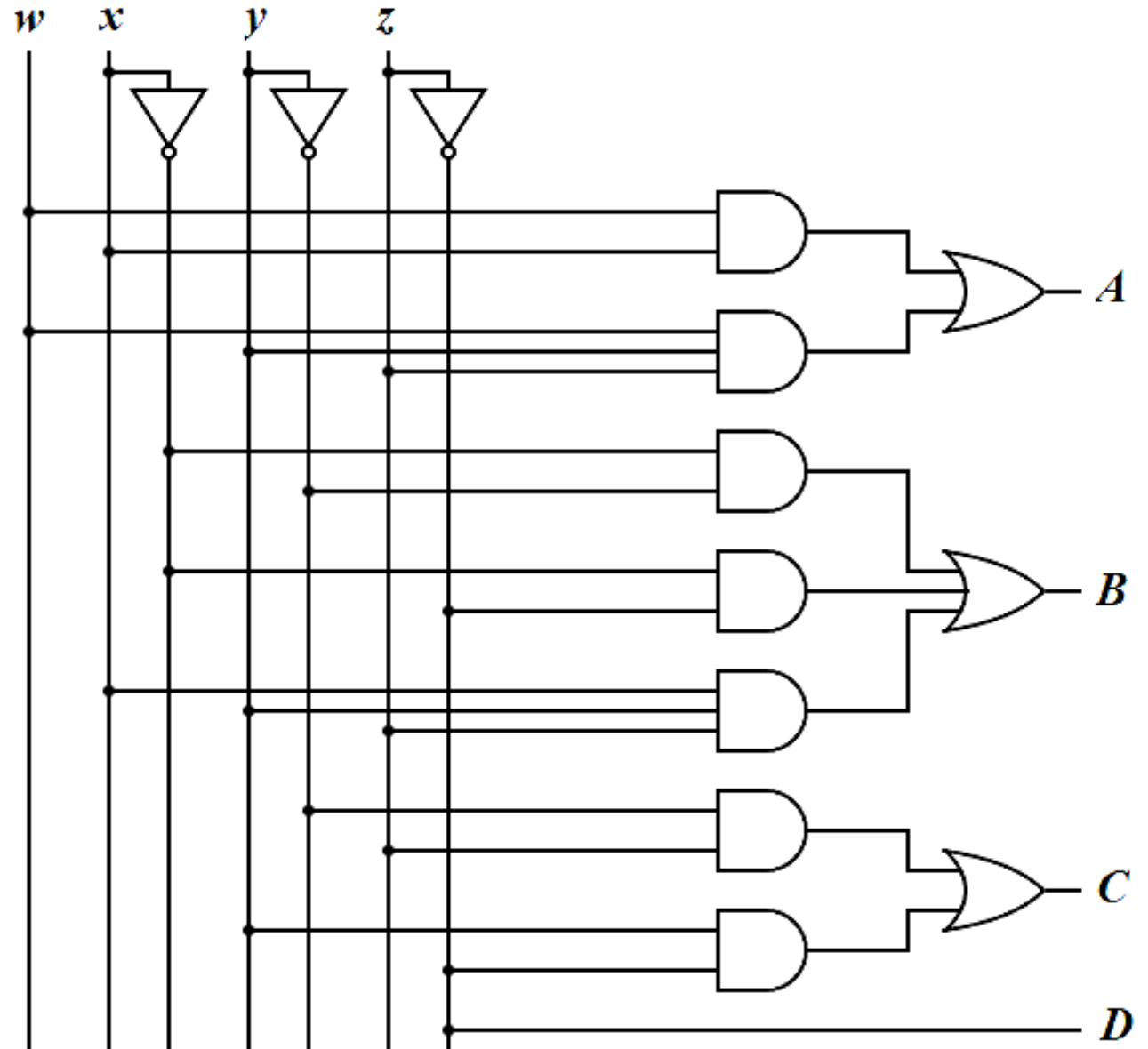
Step-4: Circuit Design

$$A = wx + wyz$$

$$B = x'y' + x'z' + xyz$$

$$C = y'z + yz'$$

$$D = z'$$



Thank You